

# ALMOST EQUIVALENT: MEASURING WHEN DOWNSCALED IMAGES CARRY THE FLOW-MATCHING TRAINING SIGNAL

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## ABSTRACT

Scale-wise diffusion schemes rest on a spectral argument: flow-matching noising progressively masks high spatial frequencies, so above some noise level a down-scaled latent carries *almost* the full surviving signal. We ask whether this licenses a training-side substitution — can the gradient of a low-rank adapter be computed from a downscaled latent at high noise without changing what is learned? We measure per-noise-level gradient agreement between native and downscaled latents on a DiT-based flow-matching model, with a fresh-noise floor and a VAE re-encoding control, and find that the spectral account fails quantitatively at the gradient level: the measured safety crossover sits at  $\sigma \approx 0.5$ ,  $3.5\times$  above the spectral prediction, and a  $2\times$  downscale is never gradient-safe at any noise level — a gap of 0.3 persists even at  $\sigma = 0.94$ , where the input is 97% noise by power. The reason is structural: expected gradients average noise out rather than being masked by it, and the flow-matching target  $v = \epsilon - x$  carries the clean image at unit amplitude at every  $\sigma$ . The gap decomposes into two mechanistically distinct terms: a noise-dependent input term whose amplitude is governed by the *downscale ratio*, and a  $\sigma$ -independent floor governed by the *absolute token count* of the downscaled grid, concentrated in early blocks, and further separable — by a positional-interpolation intervention — into a rotary-position-embedding share and an irreducible graph residue. The practical output is a measured (route,  $\sigma$ ) safety map and an opt-in  $\sigma$ -conditional trainer that draws the noise level first and substitutes the downscaled latent only on measured-safe steps, for a  $\sim 14\%$  wall-clock reduction at fixed step count on our corpus.

## 1 INTRODUCTION

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. A family of recent methods therefore runs part of the computation at reduced resolution, justified by a spectral observation about the forward noising process: under flow-matching noising  $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ , the per-frequency signal-to-noise ratio falls with  $\sigma$ , and high spatial frequencies drop below the noise floor first. Above a crossover noise level, a downscaled latent is claimed to carry *almost* the full surviving signal — the premise behind scale-wise distillation (Starodubcev et al., 2025), pyramidal flow matching (Jin et al., 2024), and progressive-resolution training schedules more broadly (Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023).

These methods change the *inference* trajectory or the model itself. We ask a narrower, training-side question: at high noise, can the *gradient* of an adapter be computed on a downscaled latent as a drop-in substitute for the native one — same model, same objective, same  $\sigma$  distribution, only the spatial grid changed on some steps? If the spectral account governed, the answer would be yes above the spectral crossover  $\sigma^* \approx 0.14$  for our latents, and the payoff would be large: on our corpus roughly half of all training draws sit at  $\sigma > 0.5$ .

We test this directly. Our instrument computes, per noise-level bin, the cosine between adapter gradients from native and downscaled latents, calibrated by a fresh-noise floor (how much two native gradient estimates disagree anyway) and a VAE re-encoding control (the cost of a decode–encode round trip with no resolution change). The headline results:

1. **The spectral prediction fails at the gradient level** (§4). The measured crossover for the mildest downscale ( $0.875\times$  edge) is  $\sigma \approx 0.5$ , not the closed-form spectral prediction 0.136; a  $2\times$  downscale never becomes safe at any  $\sigma$ , retaining a gap of  $\sim 0.3$  at  $\sigma = 0.94$  where the input is essentially pure noise. Spectral sufficiency of the *input* licenses representability, not gradient equivalence — and we argue the divergence is structural, not incidental: the trained quantity  $\mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$  averages noise out rather than being masked by it, and the flow-matching target  $v = \epsilon - x$  carries the clean image at coefficient one at every  $\sigma$  (§4.1).
2. **The gap has a two-term anatomy** (§5):  $\text{gap}_e(\sigma) \approx A_e m(\sigma)/G(\sigma)^p + \text{Floor}_e$ . The residual-mismatch shape  $m(\sigma)$  is *route-uniform* — all route identity lives in the network’s input-output Jacobian, not in the prediction error. The input-term amplitude  $A_e$  is governed by the downscale *ratio* (an iso-severity probe at  $1.6\times$  different absolute size reproduces the same curve), while the  $\sigma$ -independent floor  $\text{Floor}_e$  is governed by *absolute target token count*, concentrates in early blocks, and decomposes — via an exact positional-interpolation intervention on the rotary embedding — into a coordinate-system share  $\text{RoPE}_e$  (most of the mild-route floor) and a non-positional graph residue  $\text{Resid}_e$  (the bulk of the harsh-route floor). The positional share is removable only in the noise-dominated regime; with image content in the input, the stretched coordinates are off-manifold and *add* error, so it is a mechanism finding, not a training lever.
3. **A measured safety map and a practical trainer** (§6–7). Safety is a per-route boundary  $\sigma^*(\text{route})$ , not a universal ratio rule:  $\{1024 \rightarrow 896 \text{ at } \sigma > 0.5; 1280 \rightarrow 1024 \text{ at } \sigma \gtrsim 0.75\}$ , while  $896 \rightarrow 768$  and every  $\rightarrow 512$  route are never safe. We wire the measured-safe route into an opt-in  $\sigma$ -conditional trainer — draw  $\sigma$  first, fetch the downsampled latent only when the whole batch is above the gate — for a  $\sim 14\%$  wall-clock reduction at a fixed step budget, with a  $\sigma$ -gated banded rotary alignment as a measured refinement.

Everything here is deliberately observational-first: verdict criteria were frozen before each probe ran, and each mechanistic claim is grounded in a designated falsification test (several of which fired and killed proposed routes; we report those too).

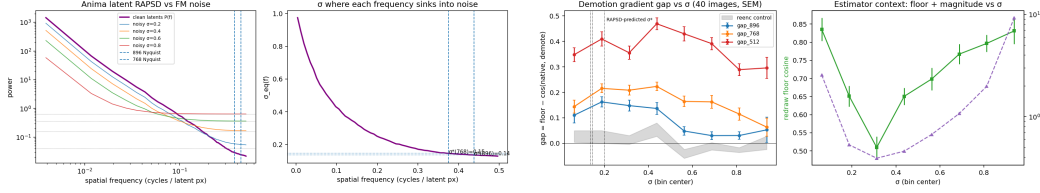
## 2 RELATED WORK

**Scale-wise and progressive-resolution diffusion.** Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogetboom et al., 2023) train at increasing resolution as a schedule. Pyramidal flow matching (Jin et al., 2024) and scale-wise distillation (SwD) (Starodubcev et al., 2025) tie resolution to the noise level itself, with SwD giving the spectral analysis we take as the hypothesis under test: their Fig. 1-style argument shows the noised downsampled latent is close in distribution to a downsampled noised latent above a crossover  $\sigma$ . Our contribution is orthogonal to all of these: we measure whether that input-level closeness transfers to the *training gradient*, holding model and objective fixed — and find it does not transfer at the predicted scale.

**Flow matching and DiT.** We work in the flow-matching / rectified-flow formulation (Lipman et al., 2023; Liu et al., 2023; Esser et al., 2024) on a Diffusion Transformer (Peebles & Xie, 2023) with rotary position embeddings (Su et al., 2024). Adapters are LoRA (Hu et al., 2022).

**Positional extension.** Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* — matching the downsampled grid’s relative phase geometry to native — to decompose the resolution-sensitivity floor, and adopt the  $\sigma$ -gated band schedule of Zhao et al. (2026) (functional form only; their spectral scale laws mispredict our measured boundaries, consistent with our main finding) as a training-time refinement.

**Gradient-agreement methodology.** Reading training decisions off gradient cosines against a noise-redraw floor follows the general logic of gradient-noise-scale analyses (McCandlish et al., 2018); our estimator operates per  $\sigma$ -bin with means over images, the regime we found reliable (split-half reliability 0.73–0.88), rather than per-image rankings, which were pure noise in our earlier attempts.



(a) Latent RAPSD and closed-form crossover  $\sigma_{eq}(f)$ . (b) Measured demotion gap per  $\sigma$ -bin (1024-tier na-Predicted  $\sigma^*$ : 0.136 / 0.146 /  $\sim 0.20$  for demotion to 896 / 768 / 512. The 896 route floors at  $\sigma \approx 0.5$ ; 512 never / 768 / 512.

Figure 1: The spectral prediction (left) versus the gradient measurement (right). The crossover is off by  $\sim 3.5\times$  and the  $2\times$  downscale never becomes safe.

### 3 SETUP AND INSTRUMENT

**Model and data.** All measurements use Anima, an open DiT-based flow-matching text-to-image model (28 transformer blocks, 3D rotary position embeddings, Qwen-family VAE), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length  $e \in \{512, 768, 896, 1024, 1280\}$ ; a tier- $e$  image occupies roughly  $(e/16)^2$  latent-patch tokens (e.g.  $\sim 4,100$  tokens at 1024,  $\sim 3,000$  at 896,  $\sim 2,160$  at 768,  $\sim 1,000$  at 512). The probe adapter is a LoRA soup checkpoint trained at native tiers; we return to this operating-point caveat in §8.

**Objective and noising.** With clean latent  $x$ , noise  $\epsilon \sim \mathcal{N}(0, I)$ , and  $\sigma \in (0, 1]$ , the noised input is  $z_\sigma = (1 - \sigma)x + \sigma\epsilon$  and the flow-matching target is  $v = \epsilon - x$ ; the loss is  $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$  with gradients taken with respect to adapter parameters only.

**Demotion.** A demotion  $e_{\text{nat}} \rightarrow e$  replaces the native latent with one produced by pixel-space downscaling of the source image to the tier- $e$  bucket followed by VAE re-encoding — the ordering SwD validated (“strategy B”); we never downsample in latent space. The caption, the noise draw structure, and the  $\sigma$  under test are held fixed; only the spatial grid changes.

**The gradient probe.** For each image and each of  $B$  uniform  $\sigma$ -bins we accumulate adapter gradients over  $D$  stratified noise draws per arm (shipped runs:  $8 \times 8$  unless noted;  $N = 40$  images) and compare arms by cosine similarity of the flattened accumulated gradients:

- **floor**: two independent draw sets at native resolution — the “how much do gradients differ anyway” null;
- **reenc**: decode–re-encode at native resolution — isolates the VAE round-trip cost that demotion necessarily pays;
- **demote arms**: one per edge  $e$  under test.

The verdict quantity is the *demotion gap*

$$\text{gap}_e(\sigma) = \cos_{\text{floor}}(\sigma) - \cos(g_{\text{nat}}(\sigma), g_e(\sigma)), \quad (1)$$

in cosine units (a mismatch *fraction*, scale-free; bin-mean SEM  $\sim 0.02$  at  $N=40$ ). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with  $\|g\|$  across bins. The re-encoding control  $\text{gap}_{\text{reenc}} \approx 0$  (band  $\pm 0.04$ ) is the instrument-validity witness in every run, and every bin-mean curve must pass a split-half reliability check over images — a discipline forced on us by an earlier failure in which per-image rankings from the same gradients had split-half reliability indistinguishable from zero. Uniform  $\sigma$ -bins make the training-time  $\sigma$ -density irrelevant to the map; density-weighting the bins reproduces our earlier pooled (marginalized) measurements as a consistency check.

Table 1: Demotion gap by  $\sigma$ -bin, 1024-tier natives ( $N=40$ , bin-mean SEM  $\sim 0.02$ , re-encoding control  $|\text{gap}_{\text{reenc}}| \leq 0.054$  everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. The spectral account predicts collapse above the second bin for every route.

| $\sigma$ bin center              | 0.06 | 0.19 | 0.31 | 0.44 | 0.56        | 0.69        | 0.81        | 0.94 |
|----------------------------------|------|------|------|------|-------------|-------------|-------------|------|
| gap <sub>896</sub> (ratio 0.875) | .110 | .162 | .148 | .137 | <b>.048</b> | <b>.030</b> | <b>.030</b> | .053 |
| gap <sub>768</sub> (ratio 0.75)  | .144 | .216 | .208 | .223 | .164        | .163        | .115        | .063 |
| gap <sub>512</sub> (ratio 0.5)   | .348 | .410 | .355 | .469 | .430        | .391        | .289        | .296 |
| cos <sub>floor</sub>             | .84  | .65  | .51  | .65  | .70         | .77         | .80         | .83  |
| $\ g\ $ (native)                 | 2.6  | 0.5  | 0.4  | 0.5  | 0.7         | 1.1         | 2.0         | 9.3  |

#### 4 THE SPECTRAL PREDICTION FAILS AT THE GRADIENT LEVEL

**The prediction.** Under unit-variance white noise the per-frequency crossover has the closed form  $\sigma_{\text{eq}}(f) = \sqrt{P(f)} / (1 + \sqrt{P(f)})$ , where  $P$  is the clean-latent radially averaged power spectral density. Anima’s VAE latents are high-frequency-quiet ( $P(f) < 1$  above  $f \approx 0.16$  cycles/latent-pixel; latent variance 0.42), giving  $\sigma^*(896) = 0.136$ ,  $\sigma^*(768) = 0.146$ ,  $\sigma^*(512) \approx 0.20$ , with tight per-image spread (p10–p90  $\approx 0.11$ –0.16): the spectral crossover is image-generic, and if it governed, demotion would be safe over almost the entire  $\sigma$  axis (Fig. 1a).

**The measurement.** Table 1 gives the measured gap curves for 1024-tier natives demoted to 896/768/512.

Three reads, against criteria frozen before the run: (i)  $\sigma$ -dependence is real and in the predicted direction (Spearman  $\sigma \rightarrow \text{gap}$ :  $-0.69/-0.57/-0.36$ ), and tier ordering holds at every bin — the *qualitative* spectral intuition survives; (ii) the crossover for the mildest route is  $\sigma \approx 0.5$ , not 0.136 — off by  $\sim 3.5\times$ ; (iii) the  $2\times$  downscale route retains gap  $\approx 0.3$  at  $\sigma = 0.94$ , where the input is  $\sim 97\%$  noise by power in every band the coarse grid cannot represent. There is no noise level at which a  $2\times$ -downscaled latent yields the native gradient.

**Ratio does not transfer.** Re-running the probe one tier down (896-tier natives, demote to 768/512) fails its pre-registered bar: 896 $\rightarrow$ 768 (ratio 0.857, nearly identical to the passing 0.875) retains a high- $\sigma$  residual of 0.06–0.12, roughly  $2\times$  the 1024 $\rightarrow$ 896 plateau and outside the re-encoding band, while the two-tiers-down control stays elevated everywhere. “One tier down at high  $\sigma$ ” is *not* the invariant; §5 shows why (the floor is governed by absolute size, and 896 $\rightarrow$ 768’s floor alone exceeds the band).

##### 4.1 WHY THE SPECTRAL ACCOUNT MUST FAIL HERE

The failure is structural, not a peculiarity of our model. Two observations, neither of which involves the spectrum of  $z_\sigma$ :

**(a) Expected gradients are never noise-masked.** “Noise power exceeds signal power in band  $f$ ” is a statement about a single sample of  $z_\sigma$ . The quantity training consumes is  $\mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$  — an expectation that averages the noise out and retains attenuated signal at *any* SNR. Input-level sufficiency licenses inference-time substitution; it never licenses training-dynamics equivalence.

**(b) The target carries the clean image at unit amplitude at every  $\sigma$ .** The flow-matching target is  $v = \epsilon - x$ . The *input* is progressively noise-masked as  $\sigma \rightarrow 1$ , but  $x$  sits in the *target* at coefficient one even at  $\sigma = 0.94$ . Writing the adapter gradient as  $g = J^T r$  (Jacobian times residual), the two grids can disagree through  $J$  itself — attention softmax over a different token count  $N$ , rotary phase density, sequence-length-dependent normalization — regardless of how little image signal the input retains.

Both observations predict exactly what we measure at  $\sigma = 0.94$ : a route-ordered gap that survives an input of essentially pure noise. The next section pins this down experimentally.

Table 2: Isolating the floor. **Endpoint:** at  $\sigma=1$  the input is exactly  $\epsilon$ , so any gap *is* the floor by construction (16 draws). **x-zero:** the image is zeroed in both input and target on every grid; any surviving gap is pure graph-shape sensitivity. Pre-registered bands in brackets; all pass, and the account’s kill switch (all endpoint gaps  $\approx 0$ ) does not fire.

| route               | endpoint gap @ $\sigma=1$ (SEM) | band         | x-zero @ $\sigma=1$ (SEM) |
|---------------------|---------------------------------|--------------|---------------------------|
| re-encoding control | −0.038 (.032)                   | [−.04, .04]  | —                         |
| 1024→896            | −0.009 (.042)                   | [0, 0.08]    | 0.004 (.018)              |
| 1024→768            | +0.127 (.054)                   | [0.04, 0.15] | 0.064 (.028)              |
| 1024→512            | +0.326 (.059)                   | [0.2, 0.4]   | 0.299 (.040)              |

## 5 ANATOMY OF THE GAP: A TWO-TERM ACCOUNT

We propose, and then test term by term, the decomposition

$$\text{gap}_e(\sigma) \approx \underbrace{A_e \cdot m(\sigma) / G(\sigma)^p}_{\text{input branch } S_1} + \underbrace{\text{Floor}_e}_{\text{graph term}}, \quad p \in [1, 2], \quad (2)$$

where  $m(\sigma)$  is the (route-uniform) mismatch of the model’s mean prediction residual across grids,  $G(\sigma) = \|g\|$  is the total gradient scale (the cosine’s implicit denominator),  $A_e$  is a  $\sigma$ -independent route amplitude, and  $\text{Floor}_e$  is a  $\sigma$ -independent term carried by the network function itself. The formula is structural rather than a quantitative fit:  $p$  is fixed only to  $[1, 2]$  by cosine geometry (2 in the small-orthogonal-mismatch limit), and  $A_e$  absorbs units. Its value is that each object is independently measurable, and that it retrodicts every curve shape and route ordering we observe. The account was pre-registered with an outcome matrix (including a kill switch) before the discriminating runs; it survived, and subsequently absorbed four refinements, each from a designated probe.

### 5.1 THE FLOOR EXISTS, IS $\sigma$ -INDEPENDENT, AND IS NOT CARRIED BY THE INPUT

Two designated probes (Table 2) establish the graph term. At the  $\sigma = 1$  *endpoint* the input is pure  $\epsilon$  — identical in distribution across arms — so the input-information term is zero by construction, and the measured gaps ( $\approx 0 / 0.13 / 0.33$  for 896/768/512) match the high- $\sigma$  plateaus of Table 1: the plateau *is* the floor. The *x-zero* probe removes the image from input *and* target everywhere; for the  $2\times$  route the surviving gap equals the endpoint gap (0.30 vs 0.33) — the floor is essentially all network-function term, with per-image content contributing nothing. Two further exonerations: the re-encoding control sits in band (the VAE round trip is not the cause), and a forward-only probe of the model’s caption-conditioned *prior* across grids shows route distances that are flat where the floors are strongly ordered — so the floor’s route-ordering lives in the Jacobian factor  $J$ , not in the model’s resolution-conditioned prior  $\hat{x}_{\text{prior}}$ , and not in a training-distribution discontinuity (the never-trained 1280 grid is no more distant from 1024 than heavily-trained adjacent pairs are from each other).

### 5.2 THE INPUT BRANCH: A ROUTE-UNIFORM NUMERATOR OVER A U-SHAPED DENOMINATOR

**$m(\sigma)$  is universal.** Measuring the mean prediction residual  $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$  per grid and  $\sigma$  — the exact  $r$  in  $g = J^T r$  — shows a strong, monotone  $\sigma$ -shape (cross-grid excess  $0.89 \rightarrow 0.36$  in relative  $L_2$  from  $\sigma=0.125$  to 1) that is the *same for every route* within  $\pm 0.02$ , across routes whose gradient floors span 0 to 0.3 and whose crossovers span 0.5 to nonexistent. All route identity therefore lives in  $J$ , not in the prediction error — a pre-registered null that would have falsified the account had the residual curves been route-ordered. The decay is smooth, Wiener-like posterior shrinkage with no hard gate at the spectral crossover: gap curves read as “collapsed” where  $S_1$  sinks below the instrument band, not at  $\sigma_{\text{eq}}$ .

**$G(\sigma)$  shapes the curve.** Because the gap is a cosine it measures a mismatch *fraction*; the total gradient norm is U-shaped in  $\sigma$  ( $2.6 \rightarrow 0.4$  at  $\sigma \approx 0.3 \rightarrow 9.3$ ; Table 1, last row) — large at high  $\sigma$  where the model is pulling composition out of the prior and the residual is image-scale, small in the mid- $\sigma$  refinement regime, and inflated again at low  $\sigma$  by irreducible  $\epsilon$ . A monotone absolute mismatch divided by a U-shaped norm yields exactly the measured interior-peaked gap curves: bin-mean gap tracks  $1/\|g\|$  (Spearman +0.55 to +0.90 across arms and runs, and within images), and

Table 3: Iso-severity probe: gap by  $\sigma$ -bin on the 1280-tier cache ( $N=24$ , 4 draws/bin) against the corpus 1024 $\rightarrow$ 896 curve. The ratio-matched routes coincide within  $\sim 1.4$  combined SEM at every bin despite a  $1.6\times$  difference in absolute target size; the same-native harsher-ratio route separates. \*896’s canonical 16-draw endpoint is  $-0.009$  (Table 2).

| $\sigma$                             | 0.125 | 0.375 | 0.625 | 0.875 | 1.0   |
|--------------------------------------|-------|-------|-------|-------|-------|
| re-encoding control                  | +.047 | −.005 | +.048 | +.020 | −.012 |
| 1280 $\rightarrow$ 1120 (ratio .875) | .129  | .110  | .054  | −.012 | −.007 |
| 1280 $\rightarrow$ 1024 (ratio .80)  | .170  | .178  | .111  | .002  | .061  |
| 1024 $\rightarrow$ 896 (ratio .875)  | .132  | .076  | .077  | .049  | .100* |

renormalizing by  $G$  flips the anomalous low- $\sigma$  dip into the monotone maximum in every arm of both runs. This is a post-hoc consistency analysis (flagged as such): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

### 5.3 THE GOVERNORS: RATIO SETS THE AMPLITUDE, ABSOLUTE SIZE SETS THE FLOOR

With the floor isolated, what orders the routes? Two probes give a clean division of labor:

**Absolute size, not ratio, sets the floor.** The route 1280 $\rightarrow$ 1024 is *more* aggressive by ratio (0.80) than the never-safe 896 $\rightarrow$ 768 (0.857), yet its floor is  $\approx 0$  and it enters the re-encoding band at  $\sigma \gtrsim 0.75$  — refuting any pure ratio rule for the floor. Conversely the floor grows monotonically as the *target* grid shrinks (0/0.06–0.13/ $\sim 0.3$  at target  $\sim 3,000/2,160/1,000$  tokens), consistent with “floor = how well the coarse graph approximates the fine graph’s computation.”

**Ratio, not absolute size, sets the amplitude  $A_e$ .** A synthetic iso-severity route 1280 $\rightarrow$ 1120 — edge ratio 0.875 exactly matched to 1024 $\rightarrow$ 896, but  $1.6\times$  its absolute target size — reproduces the 1024 $\rightarrow$ 896 curve bin-for-bin and its crossover  $\sigma^* \approx 0.5$ , while the same-native 1280 $\rightarrow$ 1024 (ratio 0.80) separates and floors later (Table 3). Amplitude follows ratio; with  $\text{Floor}_{1120} \approx 0$  this completes the division: *ratio sets  $A_e$* ; *absolute target capacity sets  $\text{Floor}_e$* . The pre-registered contrast (ratio-governed  $\Rightarrow \sigma^* \approx 0.5$ ; capacity-governed  $\Rightarrow$  earlier floor) discriminated cleanly.

### 5.4 WHERE THE FLOOR LIVES: DEPTH, AND A POSITIONAL DECOMPOSITION

**Depth, not module type.** Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) shows the floor concentrated in *depth*: early blocks (0–9, peaking at 3–8) carry  $\sim 3\times$  the late-block gap, uniformly across every module type within a block (at  $512/\sigma=1$ , every type sits in 0.22–0.31). The content share (endpoint minus  $x$ -zero) is a late-block minority effect. The mechanism picture: the first  $\sim 10$  blocks build grid-calibrated token statistics; the divergence propagates into every parameter type’s gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Notably, query and key projections show *zero* excess over value projections — which we initially read as refuting a rotary-embedding mechanism. That inference was wrong, in an instructive way.

$\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$ . Landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. The discriminating experiment is an origin-side *intervention*: evaluate the demoted grid with positionally-interpolated fractional rotary coordinates (patch  $i$  at  $i \cdot H_{\text{nat}}/H_{\text{dem}}$  per axis), making its relative phase geometry exactly match native. At the  $\sigma=1$  endpoint this *erases* the 768 floor ( $+0.080 \rightarrow -0.001$ ; paired  $\Delta = +0.081$ , 78% of images improved) and removes  $\sim 30\%$  of the 512 floor ( $+0.320 \rightarrow +0.224$ ), while leaving the in-band 896 control untouched. The mild-route floor is mostly a *coordinate-system artifact*; the harsh-route bulk is a genuine non-positional graph residue (attention-over- $N$  statistics, sequence-length-dependent normalization, capacity). The capacity governor of §5.3 attaches to  $\text{Resid}_e$ .

**The positional share is a noise-regime artifact only.** A  $\sigma$ -resolved follow-up closes the tempting route this opens (training 1024 $\rightarrow$ 768 through interpolated coordinates): with image content in the input, the stretched forward is *off-manifold* — the interpolated arm is *worse* than the plain de-

Table 4: The measured safety map.  $\sigma^*$  = noise level above which the demotion gap is within the re-encoding control band; token ratio = per-forward cost of the demoted grid. Every entry is a bin-mean read at  $N \geq 24$  with split-half reliability and in-band controls.

| route     | edge ratio  | target tokens | Floor <sub>e</sub> | $\sigma^*$           | verdict                             |
|-----------|-------------|---------------|--------------------|----------------------|-------------------------------------|
| 1024→896  | 0.875       | ~3,000        | $\approx 0$        | $\approx 0.5$        | <b>safe</b> , $\sigma > 0.5$        |
| 1280→1024 | 0.80        | ~4,100        | $\approx 0$        | $\in (0.625, 0.875)$ | <b>safe</b> , $\sigma \gtrsim 0.75$ |
| 1280→1120 | 0.875       | ~4,800        | $\approx 0$        | $\approx 0.5$        | safe (synthetic grid)               |
| 896→768   | 0.857       | ~2,160        | 0.06–0.13          | —                    | never in band                       |
| 1024→768  | 0.75        | ~2,160        | 0.06–0.13          | —                    | never in band                       |
| any →512  | $\leq 0.57$ | ~1,000        | ~0.3               | —                    | never in band                       |

moted arm through  $\sigma \in [0.56, 0.81]$  (paired  $-0.05$  to  $-0.11$ ) and wins only at  $\sigma \geq 0.94$ . The floor’s positional share is erasable only where the input is noise-dominated, so the decomposition is a mechanism finding, not a training lever — and the input-branch term of the 768 route is fatal on its own ( $+0.09$ – $0.22$  over control across the window). A frequency-*selective* banded alignment does survive in-window (§7).

## 5.5 WHAT THE ACCOUNT RETRODICTS

Taking Eq. 2 with measured  $m$ ,  $G$ ,  $A_e$ , Floor<sub>e</sub>: (i)  $\sigma^* \approx 0.5$  rather than the spectral 0.14 — the crossover is where  $A_e m/G^p$  sinks below the instrument band, a property of the ratio, and the 3.5× discrepancy is expected, not anomalous; (ii) the interior peak and low- $\sigma$  dip of every gap curve; (iii) tier ordering at every bin ( $A_e$  and Floor<sub>e</sub> are both severity-ordered); (iv) high- $\sigma$  persistence equal to the floor; (v) the 896→768 failure (its floor alone exceeds the band — no  $\sigma$ -gate can rescue it); (vi) 1280→1024 flooring *later* ( $\sigma^* \approx 0.75$ ) but cleanly (harsher ratio  $\Rightarrow$  larger  $A_e$ ; large target  $\Rightarrow$  Floor  $\approx 0$ ) — the single observation that breaks both pure-ratio and pure-capacity route-ordering stories; (vii) route-uniform residual curves under route-ordered floors, impossible if the boundary ordering were prediction-side.

A corollary worth stating plainly: the safety of 1024→896 is an *empirical smoothness statement about this network’s function across nearby token counts*, not an information-theoretic statement about the data. There was never a reason to expect a universal ratio invariant.

## 6 THE (ROUTE, $\sigma$ ) SAFETY MAP

Table 4 is the practical summary. Safety is a per-route boundary  $\sigma^*(\text{route})$ , jointly determined by ratio (through  $A_e$ ) and absolute target size (through Floor<sub>e</sub>); neither alone predicts the ordering. The batch-aggregate view is, if anything, stronger than the per-image one: summing per-image gradients before the cosine — the object SGD actually follows — collapses the 1024→896 gap to  $\approx 0$  at every bin  $\sigma \geq 0.625$ , i.e. the small per-image residual behaves as draw noise that cancels across images rather than as a shared bias. A redundancy-stratified analysis finds no per-image demotion-targeting lever (null trend across strata), consistent with the floor being an image-generic network property.

## 7 $\sigma$ -CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured — no extrapolation beyond the map:

**$\sigma$ -first draw.** The trainer draws each batch’s  $\sigma$  *before* fetching latents, from the unchanged training density (the  $\sigma$  marginal is untouched — demotion conditions on  $\sigma$ , never the reverse). When every sample in the batch is above the gate ( $\sigma > 0.5$ ), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe’s own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream — noise, target, masking, loss — derives from the swapped latent. Off-route images always train native, as does validation.

**Banded rotary alignment,  $\sigma$ -gated.** Motivated by §5.4, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the de-

moted extent, native spacing for high-frequency bands, ramp between, with the band thresholds scheduled by a sigmoid gate in  $\sigma$  following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs — it erases the static variant’s low- $\sigma$  liability ( $+0.064 \rightarrow +0.033 \pm 0.025$ ) while preserving its high- $\sigma$  gains (best and most stable arm at  $\sigma \geq 0.59$  and the endpoint, paired  $-0.05$  vs plain demotion). It is a refinement candidate, not a gate-widener: the gate stays  $\sigma > 0.5$ .

**Cost accounting.** With  $\sim 50\%$  of draws above the gate and the demoted forward at  $\sim 0.72\times$  tokens, the fixed-step epoch cost is  $\approx 0.5 \cdot 0.72 + 0.5 = 0.86$  — a  $\sim 14\%$  wall-clock ceiling on our corpus (96% of records are on the safe route), scaling with high-tier content. The pitch is wall-clock at fixed steps, never “more steps in the same time.”

**Weight-space sanity.** Under bit-exact deterministic training (paired runs; without determinism, twin runs put a chaos floor of 0.41 on checkpoint-displacement cosine, which would swamp the read), the rotary refinement adds no displacement over the demotion arm it refines (base $\leftrightarrow$ refined 0.319 vs base $\leftrightarrow$ demoted 0.320), with its footprint concentrated in the low-signal early/mid blocks — consistent with §5.4’s depth profile.

**What this section does not claim.** The end-to-end acceptance gate — a fixed-step A/B at training scale scoring generation-quality non-inferiority (CMMD; Jayasumana et al., 2024) against the realized wall-clock saving — is designed and pre-committed (a regression closes the line) but not yet run; all claims above are gradient-level and weight-space-level. We flag this deliberately: the gap between gradient-level equivalence at  $N=40$  and thousands of optimizer steps is exactly the kind of gap this paper is about.

## 8 LIMITATIONS

**One model family, one operating point.** All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint first is the designated bound on how far the map generalizes. The floor’s dependence on the fine-tuning operating point through  $J$  is untested (the designated probe: fine-tune at the 1280 tier, re-probe 1280 $\rightarrow$ 1024).

**Gradient-level, not end-task.** The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed only by the pending A/B above.

**Instrument resolution.** Bin-mean SEM  $\sim 0.02$  and the re-encoding band  $\pm 0.04$  set the floor on what “safe” can mean; residual gaps of that order are undetectable by construction. The  $\sigma^*$  boundaries are localized only to a bin.

**Scope.** Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The  $\sigma$ -gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

## 9 CONCLUSION

“Almost equivalent” input signal is not almost-equivalent training signal. The spectral noise-masking argument that justifies scale-wise *inference* schemes mispredicts the training-side substitution boundary by  $3.5\times$  and misses that a  $2\times$ -downscaled latent never yields the native gradient at any noise level — because expected gradients are not noise-masked, the flow-matching target carries the clean image at every  $\sigma$ , and the network function itself is grid-calibrated in its early blocks, partly through its positional coordinate system and partly through an irreducible dependence of the computation on token count. What replaces the spectral rule is not a cleaner formula but a measured map: per-route boundaries governed by downscale ratio on the noise-dependent term and absolute grid size on the noise-independent one. On the one route our corpus makes safe, the map converts into a free  $\sim 14\%$  of wall-clock. We offer the instrument — per- $\sigma$ -bin gradient agreement with a redraw floor and a re-encoding control, criteria frozen before data — as the cheap, general test that any future “train part of the time at lower resolution” proposal should pass before it is believed.

## REPRODUCIBILITY STATEMENT

The probe scripts (`run_sigma_probe.py`, `run_prior_distance.py`, `rapd.py`, `compare_ckpt_dw.py`), the trainer wiring (`--sigma_lowres`), frozen pre-registration documents, and every run's raw per-image records are in the public repository at [https://github.com/sorryhyun/anima\\_lora](https://github.com/sorryhyun/anima_lora) (project line `project/sigma_lowres/`). Paired training runs use a `--deterministic` mode (deterministic attention backward included) verified bit-identical across twin runs.

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## A INSTRUMENT DETAILS

**Estimator.** Per image, arm, and  $\sigma$ -bin, adapter gradients are accumulated over  $D$  stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at  $D=8$  are not comparable *across* bins (the floor drops where  $\|g\|$  is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability  $\approx 0$  and is never used.

**Controls.** The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; its gap is the validity band ( $\pm 0.04$ ; observed  $|\cdot| \leq 0.054$  across all bins of the main runs, per-module-group  $\leq 0.045$  in the split runs). The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer’s  $\sigma$ -density reproduces earlier  $\sigma$ -marginalized measurements from the same instrument family (consistency check).

**Endpoint and x-zero modes.** The  $\sigma=1$  endpoint bin feeds input exactly  $\epsilon$  (the input-information term vanishes by construction). The x-zero mode zeroes  $x$  in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the  $(1-\sigma)x$  term absent, low- $\sigma$  bins are off-manifold and the  $\sigma=1$  read is primary. In that regime the residual is  $\approx -\hat{x}_{\text{prior}}$ , so “graph-dominated” includes the model’s grid-conditioned prior; the prior-distance probe (§5.1) subsequently dissociated the prior from the floor’s route ordering.

## B ADDITIONAL TABLES

Table 5: Ratio-transfer run: 896-tier natives ( $N=40$ ), pre-registered bar “gap<sub>768</sub> within the re-encoding band at  $\sigma \geq 0.5$ ” — **fails** (residual 0.06–0.12,  $\sim 2\times$  the 1024→896 plateau).

| $\sigma$ bin | 0.06 | 0.19 | 0.31 | 0.44 | 0.56 | 0.69 | 0.81 | 0.94 |
|--------------|------|------|------|------|------|------|------|------|
| 896→768      | .114 | .209 | .168 | .143 | .124 | .056 | .092 | .061 |
| 896→512      | .318 | .372 | .308 | .340 | .320 | .280 | .329 | .217 |

Table 6: Route-uniformity of the mean prediction residual: cross-grid excess of  $\bar{r} = \mathbb{E}_{\epsilon} [\hat{v} - (\epsilon - x)]$  over split-half floors (relative  $L_2$ ; same-grid re-encoding control  $\leq 0.02$  everywhere). The three main routes coincide within  $\sim \pm 0.02$  at every  $\sigma$  while their gradient floors span 0 to 0.3.

| $\sigma$  | 0.125 | 0.375 | 0.625 | 0.875 | 1.0  |
|-----------|-------|-------|-------|-------|------|
| 1280→1024 | .885  | .825  | .715  | .465  | .360 |
| 1024→896  | .862  | .808  | .706  | .459  | .378 |
| 896→768   | .860  | .814  | .715  | .466  | .393 |
| 768→512   | .897  | .872  | .767  | .508  | .435 |

Table 7: Depth localization of the floor: mean per-block gap at  $\sigma=1$ , early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry  $\sim 3\times$  the late-block gap; the content share (ep – xz) is a late-block minority effect.

| route    | ep early | ep late | xz early | xz late |
|----------|----------|---------|----------|---------|
| 1024→512 | .357     | .223    | .351     | .125    |
| 1024→768 | .164     | .085    | .121     | .033    |
| 1024→896 | .023     | .015    | .044     | .012    |

Table 8: Positional-interpolation intervention at the  $\sigma=1$  endpoint ( $N=40$ , 16 draws): exact phase-geometry alignment erases the mild-route floor and  $\sim 30\%$  of the harsh-route floor.

| route              | plain (SEM)   | PI-aligned (SEM) | paired $\Delta$ (SEM) | improved |
|--------------------|---------------|------------------|-----------------------|----------|
| 1024→896 (control) | −0.021 (.041) | −0.040 (.040)    | —                     | —        |
| 1024→768           | +0.080 (.048) | −0.001 (.039)    | +0.081 (.031)         | 78%      |
| 1024→512           | +0.320 (.058) | +0.224 (.056)    | +0.096 (.039)         | 70%      |